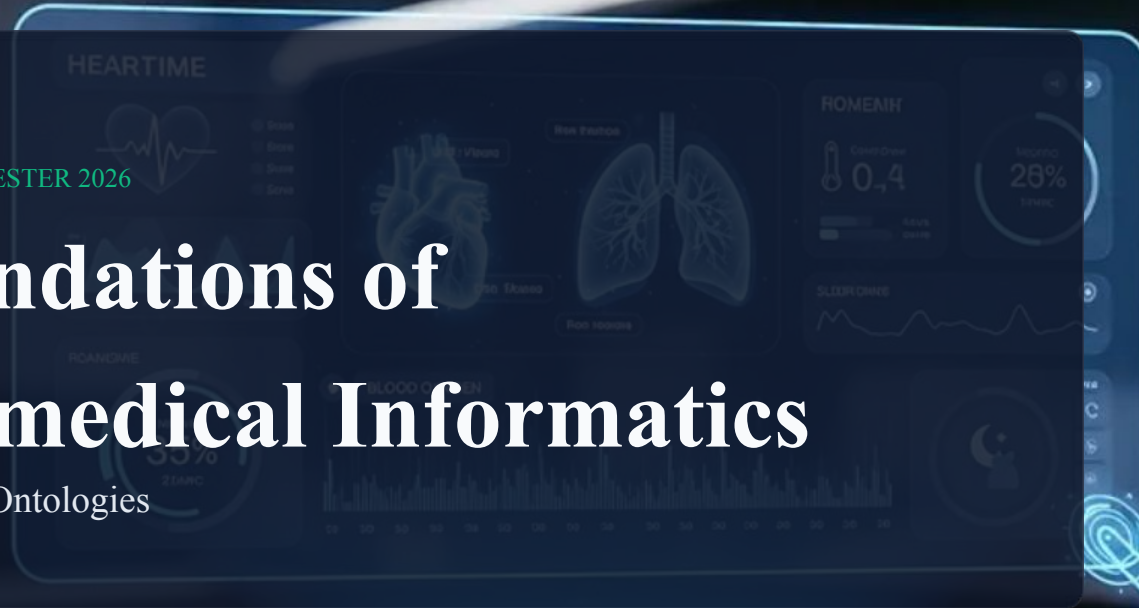
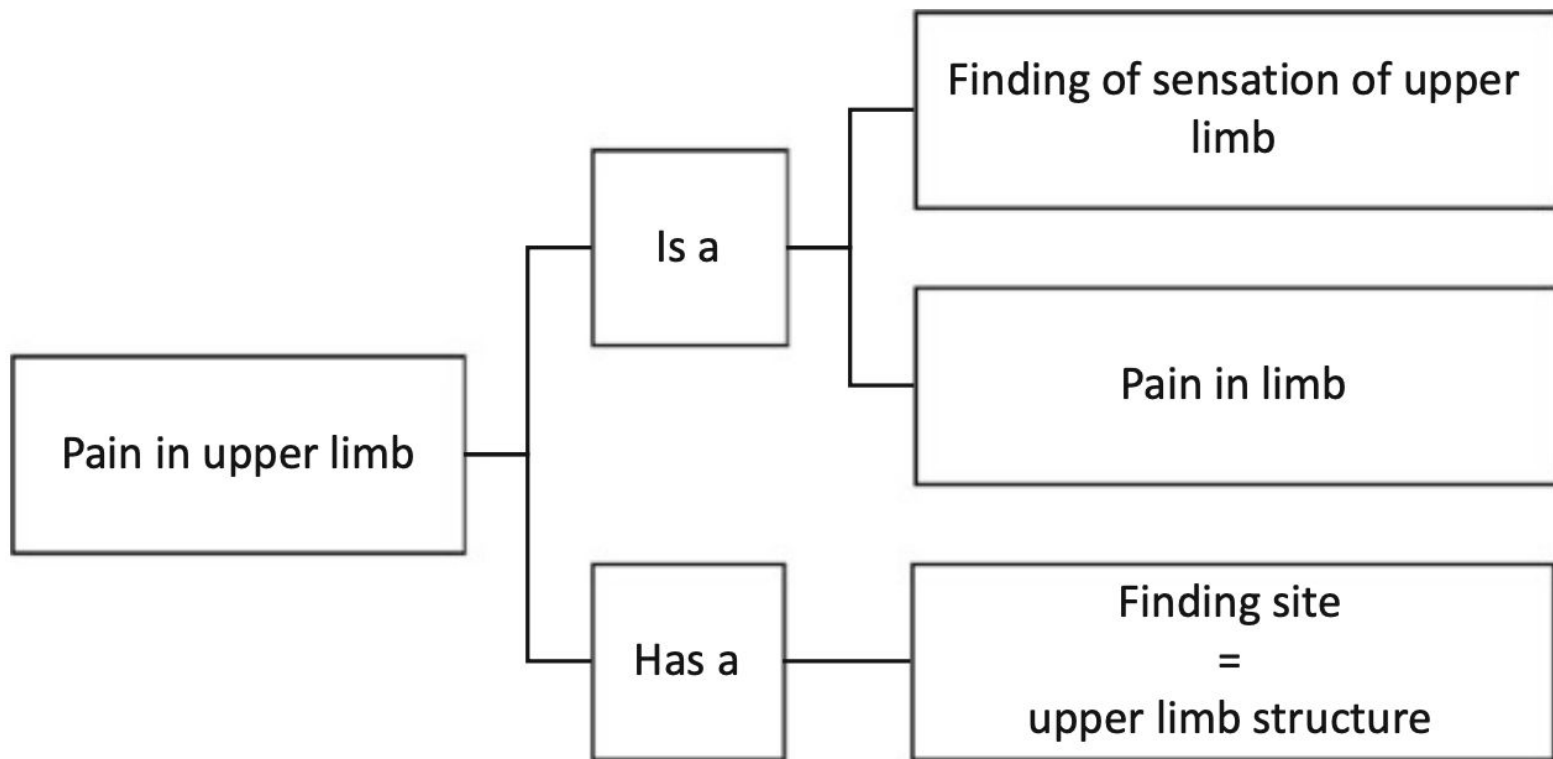


WINTER SEMESTER 2026

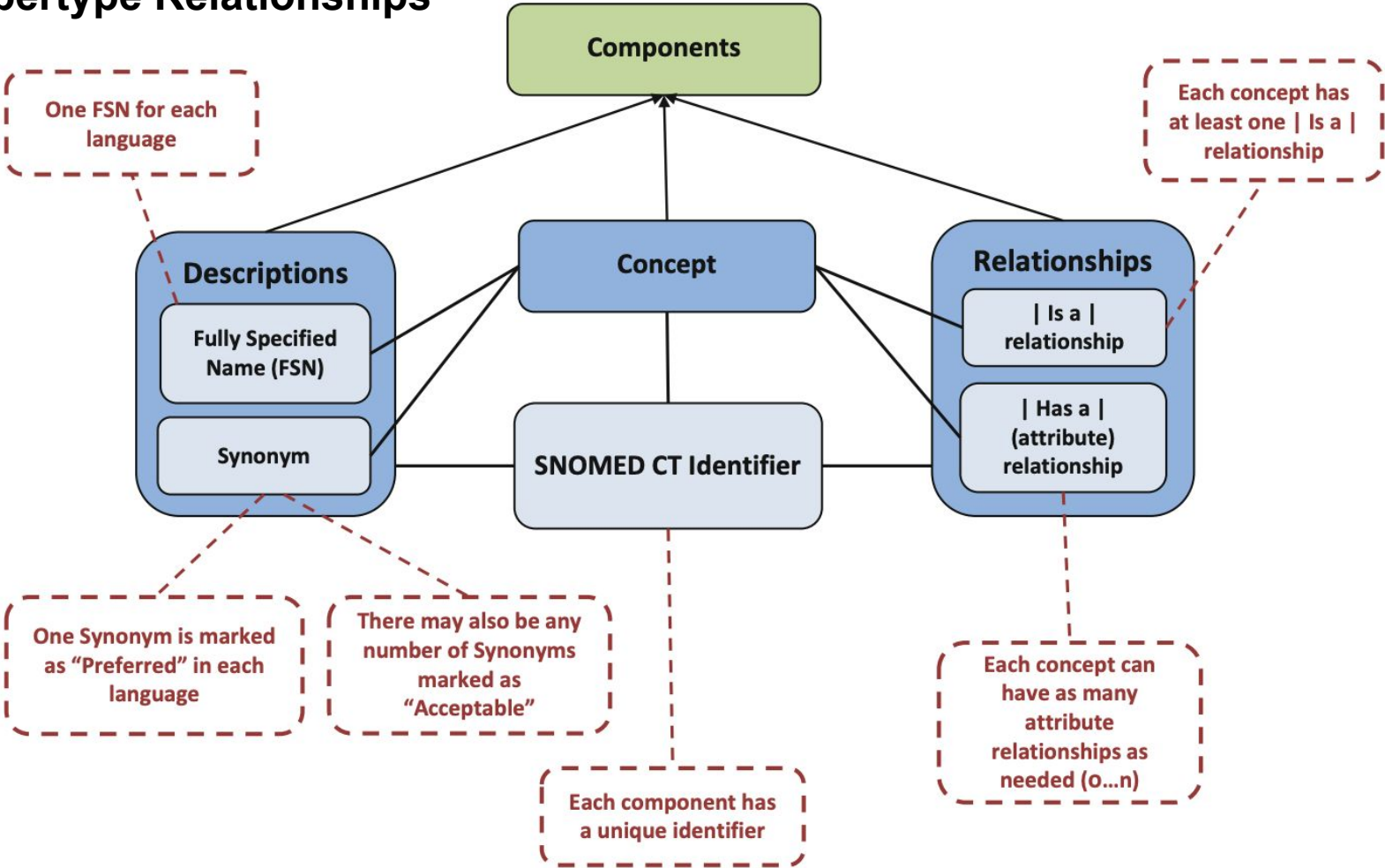
Foundations of Biomedical Informatics

Lecture 8: Ontologies





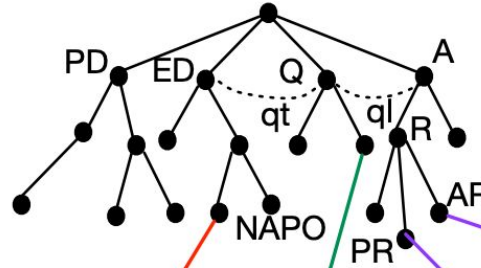
Supertype Relationships



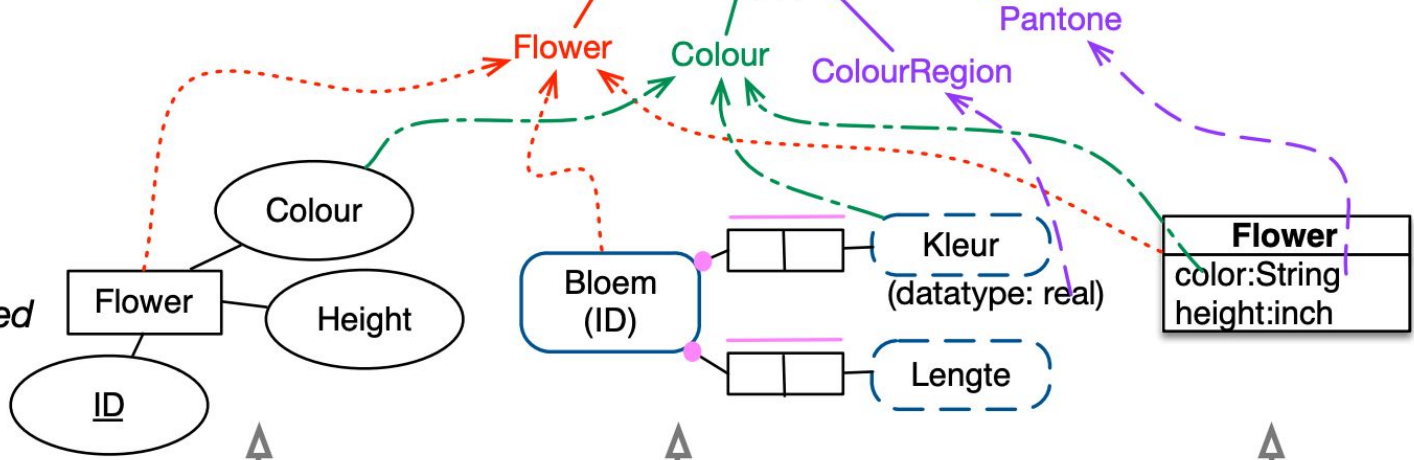
What are Ontologies?

Ontology

provides the common vocabulary and constraints that hold across the applications

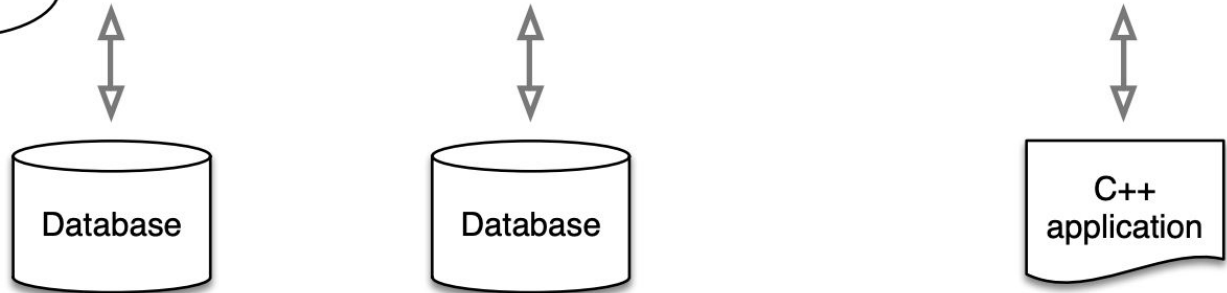


Conceptual model
shows what is stored in that particular application

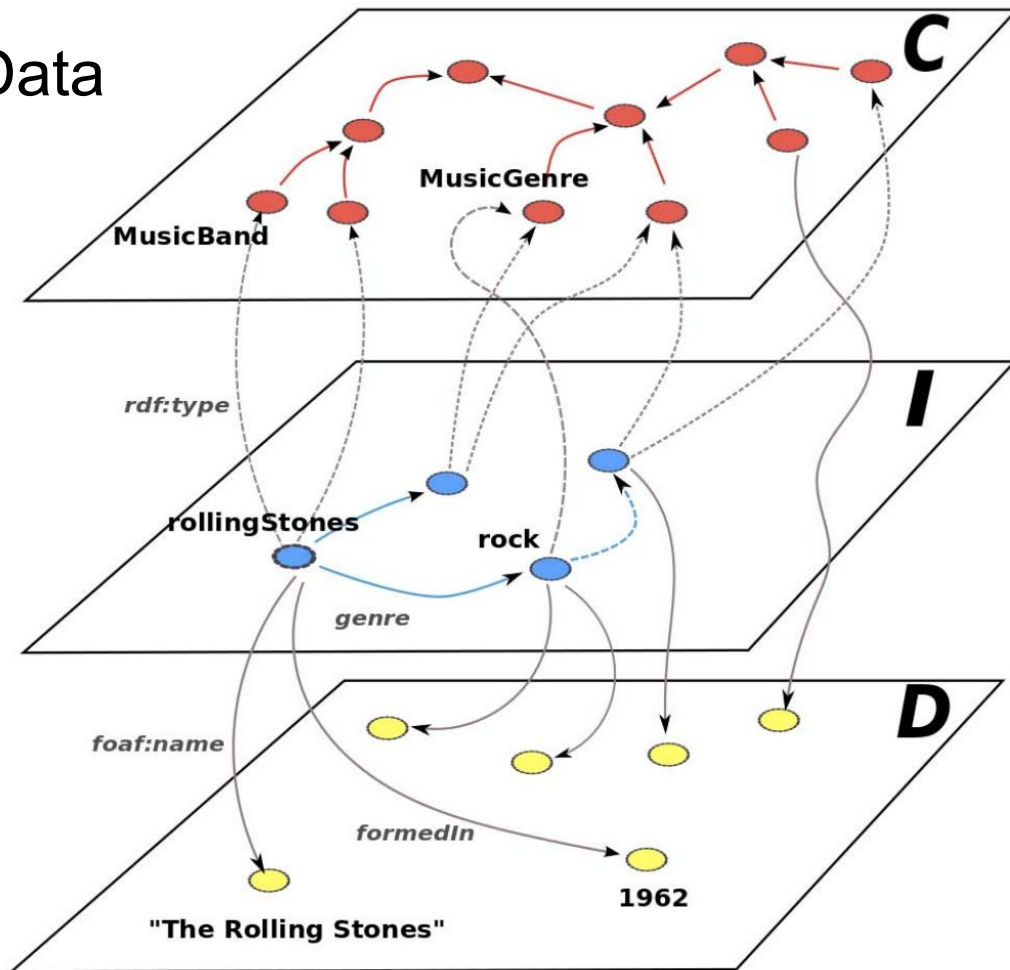


Implementation

the actual information system that stores and manipulates the data



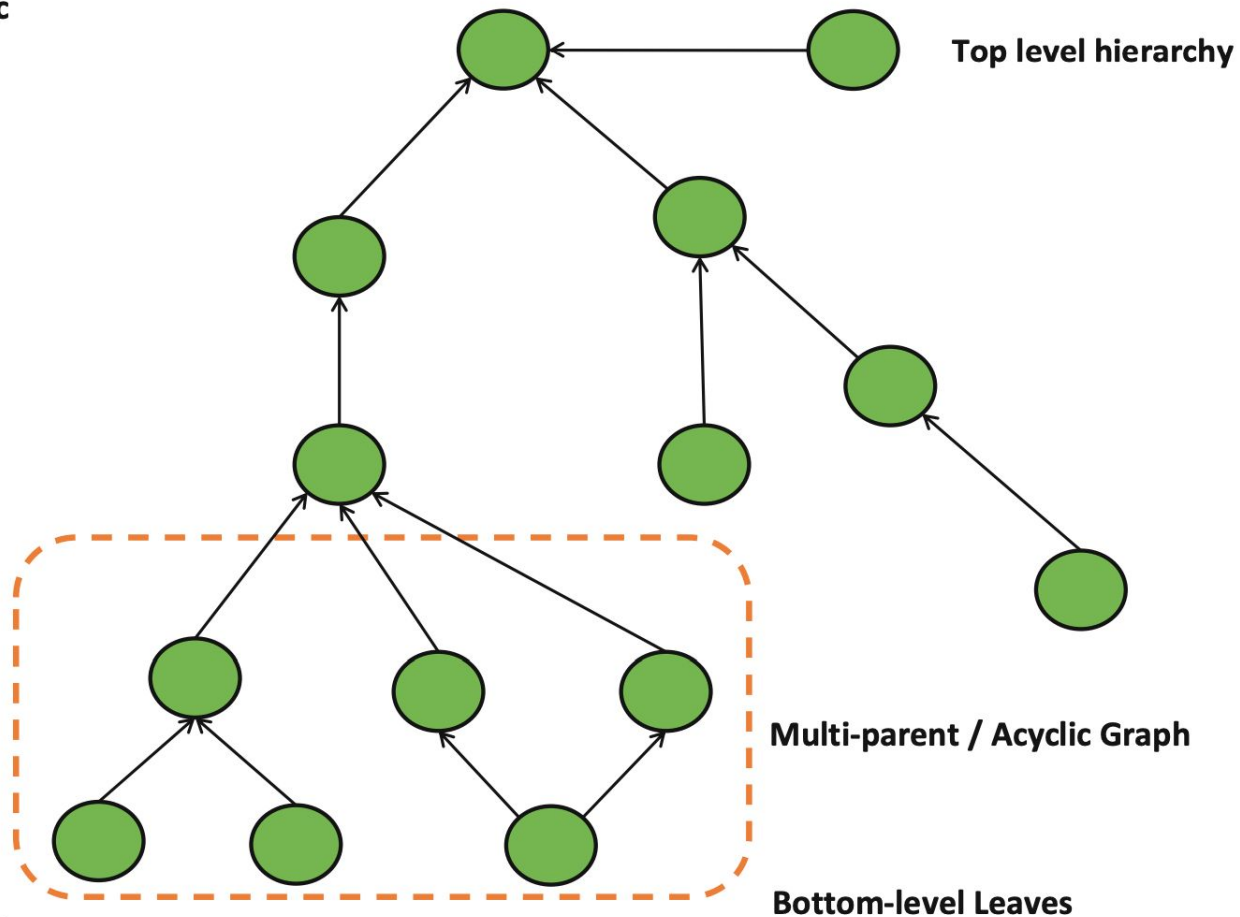
Concepts Instances Data



Generic

Degrees of Granularity

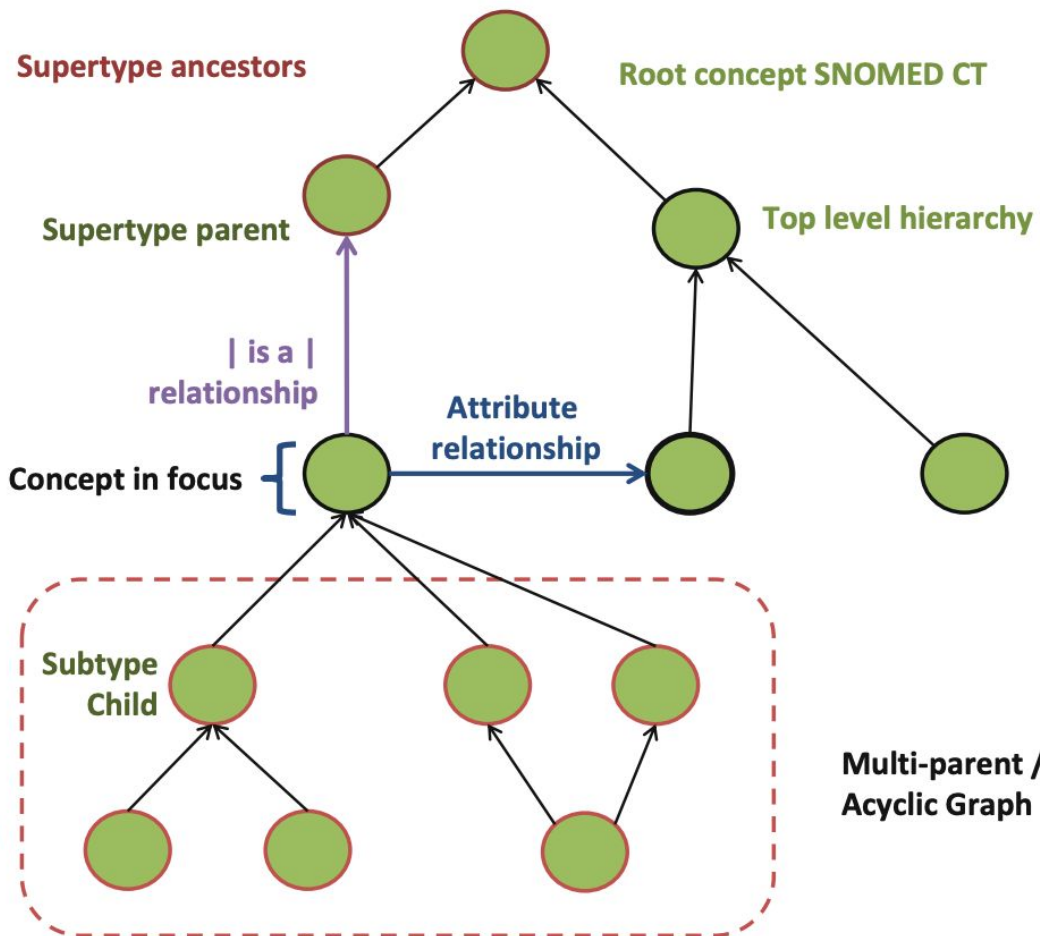
Specific



Generic

Degrees of Granularity

Specific



Defining Relationships

- (1) Supertype Relationships and (2) Defining Attributes
- Everything is a concept
- E.g. |is a| Relationship -> Concept ID: 116680003
- Every concept except the SNOMED CT root has a supertype
- A concept is **sufficiently defined** if its defining relationships are sufficient to distinguish it from all its supertype and sibling concepts.
- Another concept represented as a combination of the same defining characteristics, is equivalent to it or a subtype of it.
- Large parts of SNOMED CT are not yet sufficiently defined.
- Primitive concepts: not fully defined and do not have the unique relationships needed to distinguish them from their parent or sibling concepts.
- E.g. Pneumonia - unless defining characteristics are specified

Expressions

- Pre-coordinated: when a single concept identifier is used to represent a clinical idea
- Post-coordinated: to represent a meaning using a combination of two or more concept identifiers
- ID |Name|
- Refinement : (colon)
 - Attribute (Can be more than one, separated by comma)
 - Value

```
80146002|appendectomy|:260870009|priority|=25876001|emergency|
```

```
80146002|appendectomy|:260870009|priority|=25876001|emergency|,425391005|using access device|=86174004|laparoscope|
```

Description Logic

|concept|:|attribute|=|value|

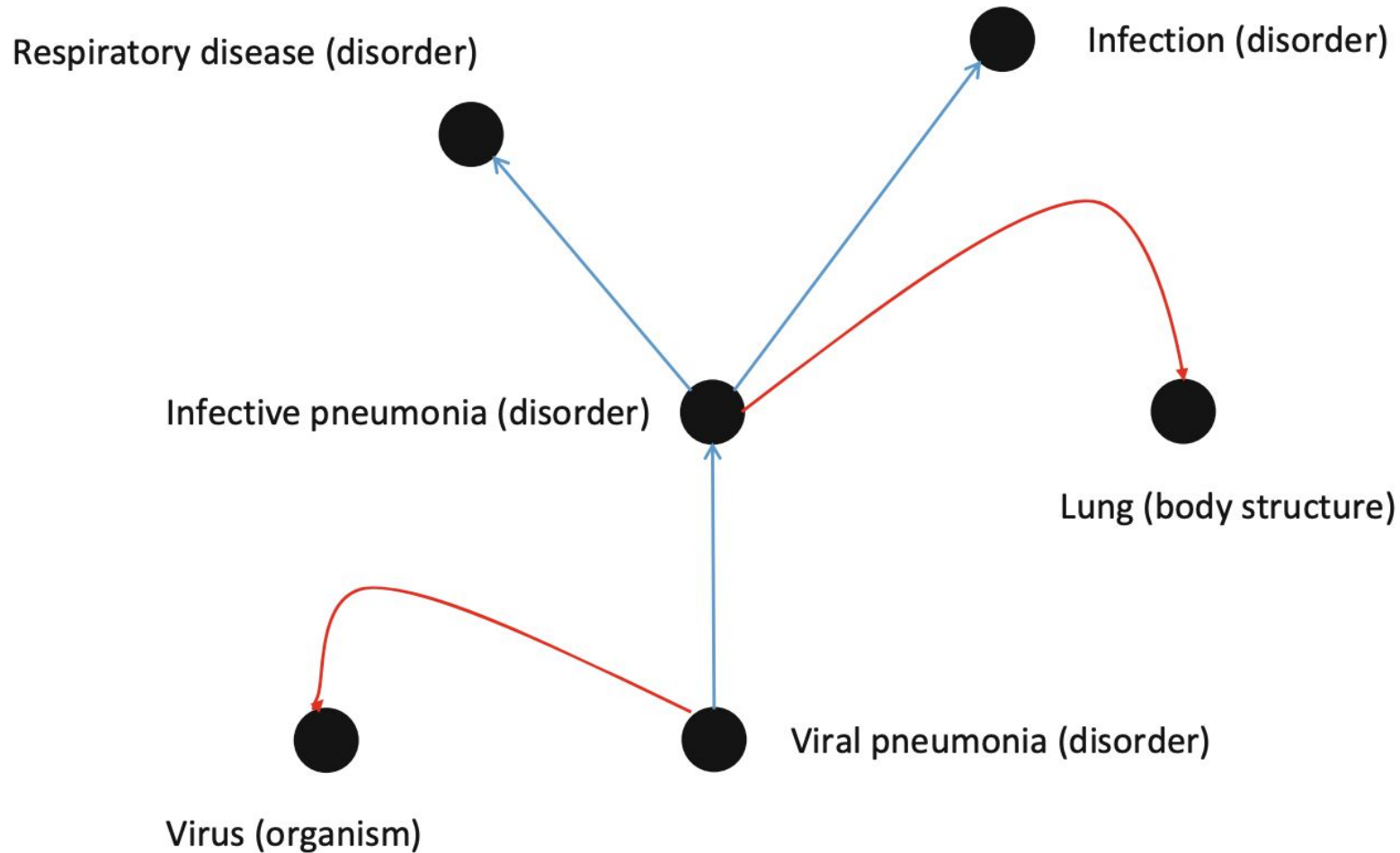
| Viral pneumonia (disorder) | is a | Infective pneumonia (disorder) |

| Viral pneumonia (disorder) | "has a" | Causative agent (attribute) |
= | Virus (organism) |

| Infective pneumonia (disorder) | is a | Respiratory disease (disorder) |

| Infective pneumonia (disorder) | is a | Infection (disorder)

| Infective pneumonia (disorder) | "has a" | Finding site (attribute) |
= | Lung (body structure) |



SNOMED CT

Last uploaded: June 10, 2022



Summary Classes Properties Notes Mappings Widgets

Jump to:

- * Body structure
- * Clinical finding
- * Environment or geographical location
- * Event
- * Observable entity
- * Organism
- * Pharmaceutical / biologic product
- * Physical force
- * Physical object
- * Procedure
- * Qualifier value
- * Action
- * Additional dosage instructions
- * Additional values
- * Alphanumeric
- * Anatomic reference point
- * Basic dose form
- * Benefits, entitlements and rights
- * Classification system
- * Clinical specialty
- * Context values
- * Descriptor
- * Disposition
- * Dose form administration method
- * Dose form intended site
- * Dose form release characteristic
- * Dose form transformation
- * Dosing instruction fragment
- * Expectations
- * Finding value
- * Findings category type
- * For profit organization
- * General clinical stage for disease AND/OR neoplasm
- * General information qualifier
- * Healthcare payer
- * Ingredient qualitative strength
- * Intellectual concepts and systems
- * Intention values
- * Legal proceedings
- * Mechanism of disease spread value
- * Mechanisms

Details Visualization Notes (0) Class Mappings (7)

Preferred Name	Hindi language
Synonyms	Hindi language (qualifier value)
ID	http://purl.bioontology.org/ontology/SNOMEDCT/161143006
Active	1
altLabel	Hindi language (qualifier value) Language Hindi
CASE SIGNIFICANCE ID	90000000000017005 90000000000020002
CTV3ID	13263
cui	C0019547
DEFINITION STATUS ID	90000000000074008
Effective time	20020131
notation	161143006
prefLabel	Hindi language
Subset member	900000000000509007--ACCEPTABILITYID--900000000000548007 900000000000508004--ACCEPTABILITYID--900000000000548007 900000000000509007--ACCEPTABILITYID--900000000000549004 900000000000497000--MAPTARGET--13263 900000000000508004--ACCEPTABILITYID--900000000000549004
tui	T171
Type ID	900000000000013009 90000000000003001
subClassOf	Indic language

<https://bioportal.bioontology.org/ontologies/SNOMEDCT?p=classes&conceptid=161143006>

ramses-antibiotics/snomedizer

R Interface to the SNOMED-CT Terminology Server REST API

[Package index](#)

Search the ramses-antibiotics/snomedizer package



Vignettes

[README.md](#)

[Functions](#) ▶

72

[Source code](#) ▶

10

[Man pages](#) ▶

12

[api_operations](#): SNOMED CT Terminology Server REST API

[concepts_descendants](#): Fetch descendants

[concepts_descriptions](#): Fetch descriptions

[concepts_find](#): Find SNOMED CT concepts

[release_version](#): Fetch SNOMED CT RF2 release

[result_completeness](#): Check that results are complete

[result_flatten](#): Flatten results from a server response

[snomed_endpoint_test](#): Test a SNOMED CT endpoint

[snomedizer_options](#): Set SNOMED CT endpoint options

snomedizer-package: snomedizer: R Interface to the SNOMED-CT Terminology Server...

In [ramses-antibiotics/snomedizer: R Interface to the SNOMED-CT Terminology Server REST API](#)

Description

SNOMED CT licensing

Licensing for reference use

Author(s)

See Also

Description

Manipulate the SNOMED CT clinical ontology using the SNOMED International Terminology Server REST API <<https://github.com/IHTSDO/snowstorm>>. 

SNOMED CT licensing

In order to use SNOMED CT, a licence is required which depends both on the country you are based in, and the purpose of your work. Although SNOMED International allows access to the SNOMED CT terminology server hosted by IHSTO to anyone for reference purpose, uses for data analysis or health care production systems is subject to other licences.

SNOMED International allows access to public SNOMED-CT terminology servers to anyone for strict 'reference purposes' under the [SNOMED International SNOMED CT Browser License Agreement](#) (see section below).

Use of SNOMED CT terminology for data analysis or health care production systems is subject to other licences. Some users are eligible for free licences:

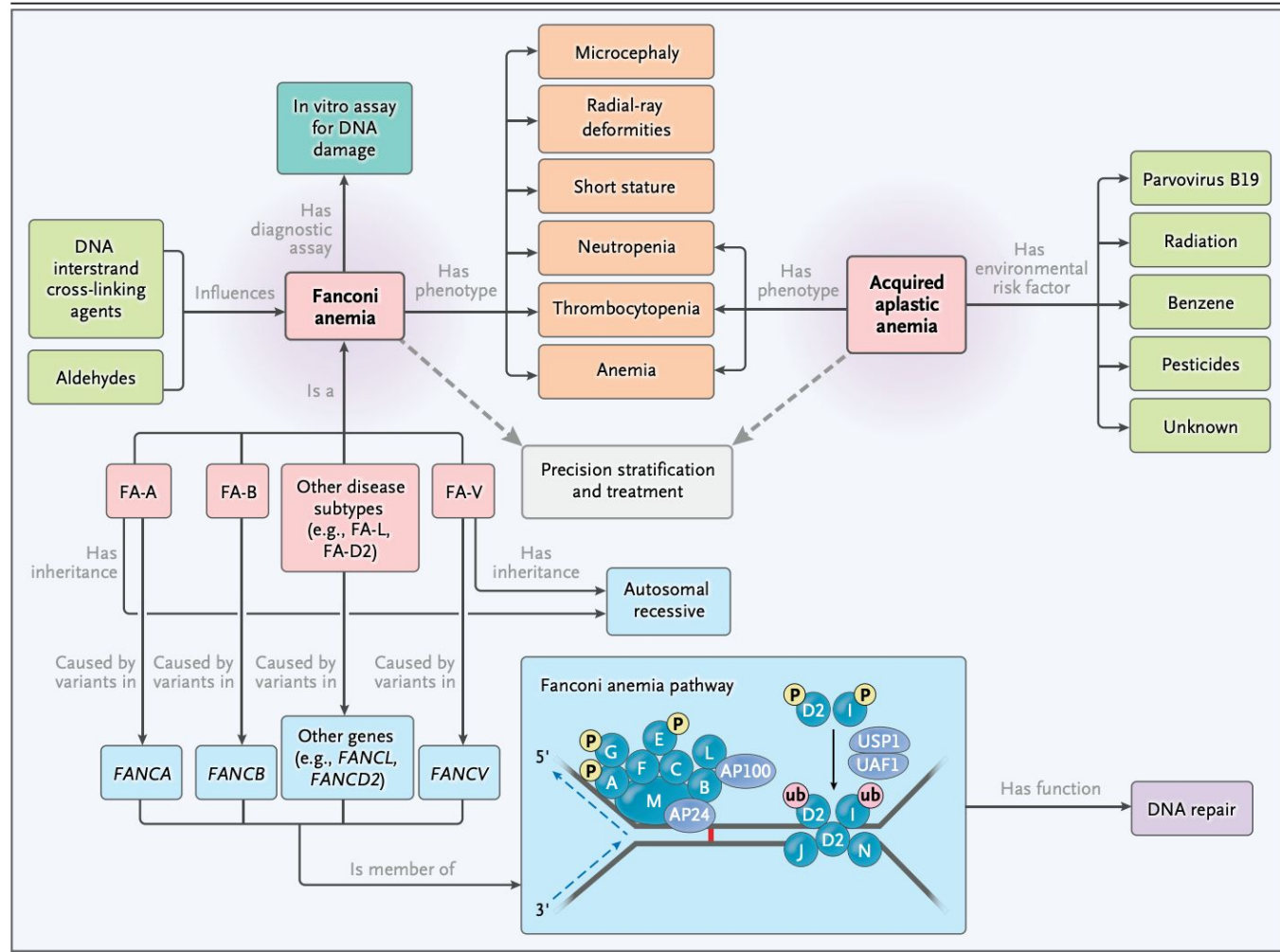
- UK-based users can obtain a licence free of charge on the [NHS TRUD website](#)
- residents of other Member Countries and low-income countries are also eligible. More information can be found on [SNOMED International's website](#).

Licensing for reference use

snomedizer contains examples and documentation that makes uses of an IHSTO public API endpoint restricted to reference uses such as testing or browsing the terminology. It is subject to the [SNOMED International Browser License Agreement](#) reproduced below:

1. The meaning of the terms "Affiliate", or "Data Analysis System", "Data Creation System", "Derivative", "End User", "Extension", "Member", "Non-Member Territory", "SNOMED CT" and "SNOMED CT Content" are as defined in the SNOMED International Affiliate License Agreement (see <http://snomed.org/license-affiliate>).
2. Information about Affiliate Licensing is available at <https://www.snomed.org/snomed-ct/get-snomed>. Individuals or organizations wishing to register as SNOMED International Affiliates can register at mlds.ihtsdotools.org, subject to acceptance of the Affiliate License Agreement (see <http://snomed.org/license-affiliate>).
3. The current list of SNOMED International Member Territories can be viewed at www.snomed.org/members. Countries not included in that list are "Non-Member Territories".
4. End Users, that do not hold an SNOMED International Affiliate License, may access SNOMED CT® using SNOMED International SNOMED CT Browser subject to acceptance of and adherence to the following sub-license limitations:
 - a) The sub-licensee is only permitted to access SNOMED CT® using this software (or service) for the purpose of exploring and evaluating the terminology.
 - b) The sub-licensee is not permitted the use of this software as part of a system that constitutes a SNOMED CT "Data Creation System" or "Data Analysis System", as defined in the SNOMED International Affiliate License. This means that the sub-licensee must not use SNOMED International SNOMED CT Browser to add or copy SNOMED CT identifiers into any type of record system, database or document.
 - c) The sub-licensee is not permitted to translate or modify SNOMED CT Content or Derivatives.
 - d) The sub-licensee is not permitted to distribute or share SNOMED CT Content or Derivatives.
5. SNOMED International Affiliates may use SNOMED International SNOMED CT Browser as part of a "Data Creation System" or "Data Analysis System" subject to the following conditions:
 - a) The SNOMED International Affiliate, using SNOMED International SNOMED CT Browser must accept full responsibility for any reporting and fees due for use or deployment of such a system in a Non-Member Territory.
 - b) The SNOMED International Affiliate must not use SNOMED International SNOMED CT Browser to access or interact with SNOMED CT in any way that is not permitted by the Affiliate License Agreement.
 - c) In the event of termination of the Affiliate License Agreement, the use of SNOMED International SNOMED CT Browser will be subject to the End User limitations noted in 4.

Automated Reasoning



SNOMED CT India Reference Sets

- Developed and released by NRCeS (India).
- Clinical domains include:
 - Tuberculosis, Dengue, Malaria
 - Stroke, Cervical cancer
 - Child health & infectious diseases
- These are production-grade clinical terminology artifacts.

Medical Subject Headings (MeSH)

- • Produced by the National Library of Medicine (NLM) since 1960
- • Used for indexing and searching biomedical literature (MEDLINE / PubMed)
- • Part of the Unified Medical Language System (UMLS)
- • Originated from Index Medicus keyword lists

- Key Properties:
 - • Controlled vocabulary
 - • Polyhierarchical structure

- Core Components:
 - • MeSH Tree Structures
 - • Annotated Alphabetic List
 - • Permuted MeSH

The 16 Trees in MeSH

- A. Anatomy
- B. Organisms
- C. Diseases
- D. Chemicals and Drugs
- E. Diagnostic & Therapeutic Techniques
- F. Psychiatry & Psychology
- G. Biological Sciences
- H. Natural Sciences
- I. Social Sciences
- J. Technology & Agriculture
- K. Humanities
- L. Information Science
- M. Named Groups
- N. Health Care
- V. Publication Characteristics
- Z. Geographicals